Collin Longoria

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Education

DigiPen Institute of Technology — Redmond, WA
Bachelor of Science in Computer Science in Real-Time Interactive Simulation, Minor in Math — *Aug. 2022 – May 2026*
3.6 GPA

Relevant Projects

Cisalpine Engine, Software Engineer | *C++, OpenGL, GLFW, Dear ImGui* — Jan. 2026 – Present

- Designed and implemented a real-time 2D Radiance Cascades global illumination system with 4-level cascade heirarchy, achieving full-scene indirect lighting at 0.8 ms per frame on RTX-class hardware, enabling dynamic light propagation from emissive elements.
- Built a GPU-driven rendering pipeline with deferred-style G-buffer extraction (color, normals, emission/opacity), multi-pass cascade merging, and ginal resolve to lightmap, supporting 10+ debug visualization modes for rapid iteration.
- Optimized compute shader dispatch using an indirect draw buffer and chunk-based spatial activation, reducing GPU workload by only processing active regions and achieving 50% frame time reduction versus native full-world dispatch.

Blok (Academic Project), Graphics Programmer | *C++, Vulkan, CUDA, GLSL* — Aug. 2025 – Present

- Developed a dual-backend voxelization and ray-tracing engine with both Vulkan-compute and CUDA-accelerated OpenGL pipelines, enabling flexible GPU performance benchmarking across APIs.
- Engineered a modular Vulkan renderer supporting data-driven pipelines and dynamic descriptor allocation, resulting in a more adaptable and extensible graphics framework.
- Optimized a high-precision mesh voxelizer converting millions of triangles into sparse voxel octrees (SVOs) in under 10 ms per frame on RTX-class hardware, supporting real-time global illumination research.

Project Mirrors (Academic Project), Gameplay Programmer | *Unreal Engine 5, C++* — Aug. 2024 – April 2025

- Developed a modular puzzle framework in C++ allowing designers to link interactive components without code, reducing technical bottlenecks in level design.
- Collaborated with technical artists to design high-performance custom shader effects and reusable Blueprint components, improving gameplay clarity and achieving project goal of 60 frames per second across all target platforms.

Elementokens (Academic Project), Engine Programmer | *C++, OpenGL, FMOD* — Aug. 2023 – April 2024

- Led engine-side development of rendering and gameplay logic, creating reusable modules for turn-based movement, combat, and dynamic map rendering to streamline feature integration.
- Designed a data-driven content pipeline that enabled game designers to prototype units, maps, and rulesets without engine changes, accelerating iteration cycles by roughly 50% .

Experience

Assistant Teacher — June 2025 – July 2025
DigiPen WANIC Program — *Redmond, WA*

- Mentored high school students in programming fundamentals, data structures, and basic game programming in JavaScript.
- Debugged student code in real-time, identifying logic errors and teaching debugging strategies.
- Developed supplemental coding exercises and mini-projects to reinforce classroom material, including interactive projects using P5.js.
- Collaborated with the lead instructor to adapt lesson content based on student feedback and performance.

Technical Skills

Languages: C++, C, CUDA, GLSL, HLSL, C#, Python, JavaScript
Engines & APIs: Unreal Engine 5, Godot, OpenGL, Vulkan, DirectX 12, WebGPU
Developer Tools: Git, Perforce, Visual Studio, VS Code, CLion, CMake, Renderdoc, NVIDIA NSight
Libraries: glslang, FMOD, Assimp, stb, GLM, SDL, GLFW, Dear ImGui